CFD Predictions and Field Measurements of NO_x Emissions From LM1600 Gas Turbine During Part Load Operation

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CFD simulations of the combustion process and formation of emissions in an industrial GE LM1600 gas turbine have been performed over a range of unit loads. Two combustion models were considered here to characterize the combustion process, the chemical equilibrium model and the nonequilibrium laminar flamelet model. The flamelet model predictions of nitric oxide concentration demonstrated much closer agreement with field measurements of continuous emission monitoring systems, due to accurate modeling of the oxygen radial concentration. The predictions made with this model are within 15 percent at maximum load and considerably better at lower loads. Field measurements also showed that nitric oxide constitutes about 95 percent of the total NO_x measured. Unburnt hydrocarbons and carbon monoxide emissions are significantly overpredicted, however, arguably as a result of neglecting their oxidation in the high temperature, fuel-lean environment of the turbine and exhaust stack downstream of the combustor.
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Introduction

Air pollution control programs in the United States increasingly require industry to continuously monitor and report emissions of pollutants, specifically oxides of nitrogen (NO_x) and sulfur (SO_x). These components are generally acknowledged as the major contributors to acid rain, while NO_x also contributes along with volatile organic compounds to the photochemical generation of ground level ozone or smog.

Current U.S. monitoring requirements are generally biased towards continuous emissions monitoring systems (CEMS) based on calibrated measurement devices. For example, the regional clean air incentives program (RECLAIM) ([1]) in southern California imposes stringent CEMS and reporting requirements for NO_x and SO_x emissions, particularly for large NO_x sources. On a federal level, the U.S. Environmental Protection Agency stipulates CEMS for installations specified under the acid rain program ([2]).

The situation is quite different in Canada, where predictive models or one-time site measurements are employed to characterize these emissions as part of the initial licensing application to the provincially based environmental regulatory bodies. Continuous monitoring of emissions is generally not a licensing requirement. Some proactive initiatives have been introduced in recent years including the Voluntary Challenge and Registry Program ([3]), which encourages private and public sector organizations across Canada to voluntarily limit and disclose their net greenhouse gas emissions. However, further regulatory changes with some form of ongoing emission monitoring could be tabled in the near future in order to meet Canada's Kyoto commitment to reduce greenhouse gases to 3 percent below 1990 levels by the year 2010.

TransCanada Transmission ([4]), which operates numerous industrial gas turbines as part of its natural gas transmission pipeline, has sponsored a program of predictive emissions monitoring systems (PEMS) in anticipation of future regulatory changes. PEMS are often based on semi-empirical expressions or one-dimensional models in an attempt to correlate emission rates to operating parameters for a given gas turbine ([5–7]). Suffice it to say, PEMS offer a very cost-effective means for estimating continuous emission levels with minimal operational and maintenance requirements, i.e., no calibration gases or expensive analyzers. A good account of the benefits of PEMS over CEMS is given by Hung ([5]), where a quantitative comparison between the two methods is provided.

As a part of this larger PEMS program, this study focuses on the ability of more sophisticated tools such as computational fluid dynamics (CFD) to accurately predict NO_x emissions. Clearly, due to the complexity and large computational requirements, CFD is not feasible for direct application as a PEMS. However, it can be used to provide some valuable data that can be implemented in PEM models.

In this paper, CFD predictions of the nitric oxide (NO) emissions from a stationary General Electric LM1600 gas turbine engine are presented. We begin with a brief review of related CFD studies of gas turbine (GT) combustion chambers. This is followed by details of the CFD modeling effort with particular emphasis on the combustion process and NO models along with the combustor geometry, grid generation, and prescribed boundary conditions. Two popular combustion models were considered in the simulations, both based on the mixture fraction probability density function (PDF) approach. The first model assumes chemical equilibrium, whereas the second laminar flamelet based model accounts for nonequilibrium effects. The results of the CFD simulations are compared with field measurements of the unit emissions taken over a range of part load operating conditions on two separate occasions.

Related CFD Studies

Many researchers have used CFD simulations to study the fundamental combustion process and heat transfer characteristics in gas turbine combustors ([8–10]). In fact, aeroengine manufacturers now commonly apply CFD during the design of new combustion chambers. Mongia ([10]) provides an excellent review of

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the application of CFD to gas turbine combustors along with some recommendations for future research from an industry perspective.

Some noteworthy CFD related NO_x studies include a series of papers with detailed experimental and numerical results for a propane-fired laboratory-scale can combustor ([11–14]). Isothermal flow simulations by Koutmos and McGuirk ([13]) demonstrated that the standard k-ε turbulence model was acceptable for predicting the velocity field despite some deficiencies in the jet-on-jet impingement region. More recently, Mueller and Knill ([14]) compared the effect of two different combustion models; namely the eddy dissipation model ([15]) and the laminar flamelet model ([16]), on predicted temperature field, species concentration and NO emissions. Their NO_x model was based on the extended Zeldovich mechanism to describe thermal NO formation along with an empirical expression by De Soete ([17]) to account for the prompt NO formation. They concluded that simple combustion models like the eddy dissipation model tend to overpredict the temperature field and NO concentrations, denoted by [NO], due to the lack of intermediate species. Conversely, the flamelet model, which better accounts for nonequilibrium effects, was much more accurate.

Further advances in CFD modeling of pollutant formation in gas turbine combustors are reported by Barths et al. ([18]). These authors introduced a series of unsteady flamelets based on a complex reaction mechanism involving 106 reactions and 14 species to describe the thermal, prompt and nitrous oxide contributions to NO_x formation and account for NO_x reburn. For n-hexane fuel, the agreement between predicted and measured NO values was excellent, within 12 percent. An analysis of the reaction paths showed that the thermal NO contributes 67 percent to the total production with the remaining 33 percent formed via the prompt mechanism. NO was globally consumed by a reburn mechanism (22 percent) and the nitrous oxide path (8 percent).

Description of CFD Model

Numerical simulations of the time-averaged turbulent flow field in the combustion chamber of the LM1600 were conducted using FLUENT 5.1 ([19]), a commercial CFD package. Calculations of the Reynolds number for the fuel, primary and secondary air injection indicate the flow conditions are fully turbulent. FLUENT solves the incompressible time-averaged governing equations for mass and momentum using a conservative control volume formulation. Closure of the Reynolds stress terms is achieved using the standard k-ε turbulence model. In spite of its inherent assumptions, i.e., isotropy, the k-ε model has been widely utilized in combustor simulations and generally yields acceptable results ([10,13,14,18]).

Mixture Fraction Approach. The mixture fraction approach was developed to model turbulent nonpremixed flames. The mixture fraction, commonly denoted by f, is defined as the atomic mass fraction that originated from the fuel stream. Under the assumptions of uniform pressure (which is valid for low speed flames), and equal turbulent species diffusivities (which is valid at high Reynolds number), the species equations can be reduced to a single equation for f. Since atomic elements are conserved during chemical reactions, the mixture fraction equation does not have a source term.

In turbulent combustion, all flow variables at a point fluctuate in time due to turbulent eddies. Turbulence is modeled in this simulation by Reynolds averaging, which may be considered as a time average of the fluctuating quantities at a point. The effects of this time varying turbulent field on the mixture fraction are accounted for by using an assumed shape for the PDF. The PDF, denoted by P, describes the proportion of time spent at each turbulent state. Hence, mean quantities, such as the mean temperature (means are denoted by a "$\sim$" overbar) are determined by a PDF weighted integration over all states,

$$\tilde{T}=\int_0^1 T(f)P(f)df. \tag{1}$$

Here, a beta PDF is used, which is specified by its first two moments, namely the mean of f and its variance. The CFD code solves equations for these first two moments, from which the mean temperatures and species can be computed according to (1)). In order to avoid the computational expense of evaluating these integrals, they are calculated a priori, tabulated and then interpolated from the table during the actual CFD simulations.

Equilibrium Model. When chemical reactions are fast relative to turbulent mixing, the chemistry may be approximated as being in chemical equilibrium. For adiabatic flames, the entire temperature and species fields are completely determined by the mixture fraction; nonadiabatic flames also depend on the enthalpy. As the residence time in the combustor decreases, however, the turbulent mixing time-scales become competitive with the chemical reaction time-scales and the chemistry deviates from thermodynamic equilibrium. The flamelet model relaxes this assumption to cases of moderate nonequilibrium.

Flamelet Model. The laminar flamelet model approximates a turbulent flame brush as an ensemble of laminar flames. Each flamelet is assumed to have the same structure as the stretched laminar flame that occurs between opposing fuel and air nozzles. For this configuration, the species and temperature in the flame can be completely described by two parameters, namely the mixture fraction, f and the scalar dissipation rate, χ. The scalar dissipation rate is a measure of the strain on the flame, and hence the extent that it departs from chemical equilibrium ([19]),

$$\chi=2D\left(\frac{df}{dx}\right)^2. \tag{2}$$

Individual laminar flamelets can be calculated from equations governing individual species transport and an overall energy balance,

$$\rho\frac{\partial Y_j}{\partial t}=\frac{1}{2}\rho\chi\frac{\partial^2 Y_j}{\partial f^2}+\dot{w}_i-\frac{1}{2}\left[\rho\chi\frac{1}{\mathrm{Le}_j^2}\frac{\partial \mathrm{Le}_j}{\partial f}+\frac{1}{2}(1-\mathrm{Le}_j)\times\left(\frac{\partial\rho\chi}{\partial f}+\rho\chi\frac{c_p}{\lambda}\frac{\partial}{\partial f}\frac{\lambda}{c_p}\right)\right]\frac{\partial Y_j}{\partial f} \tag{3}$$

$$\rho\frac{\partial T}{\partial t}=\frac{1}{2}\rho\chi\frac{\partial^2 T}{\partial f^2}+\frac{1}{2}\rho\chi\left[\frac{\partial c_p}{\partial f}+\sum_j\frac{1}{\mathrm{Le}_j}c_{p_j}\frac{\partial Y_j}{\partial f}\right]-\frac{1}{c_p}\sum_j\dot{w}_j h_j-\frac{1}{c_p}\left[4\sigma p\sum_j X_j a_j(T^4-T_b^4)\right]. \tag{4}$$

Note that the last term in the energy Eq. (4) represents heat loss from the flamelet due to radiation in the optically thin limit. The flamelet Eqs. (3) and (4) are solved over the range of scalar dissipation rates, integrated over the beta PDF according to Eq. (1) and tabulated to relate mean scalar quantities to the mixture fraction, its variance, and the scalar dissipation rate.

NO_x Model. Since NO_x occurs in trace quantities, it does not influence the flow field, and is accordingly post-processed. In this study, only NO is considered. The governing equation for the transport of NO is given by

$$\frac{\partial}{\partial t}(\rho Y_{\mathrm{NO}})+\frac{\partial}{\partial x_i}(\rho u_i Y_{\mathrm{NO}})=\frac{\partial}{\partial x_i}\left(\rho D_{\mathrm{eff}}\frac{\partial Y_{\mathrm{NO}}}{\partial x_i}\right)+\widetilde{w_{\mathrm{NO}}} \tag{5}$$

where u_i is the fluid velocity and D_{eff} is the effective (turbulent+laminar) diffusivity. The last term in Eq. (5), the mean NO source term, is computed via Eq. (1) based on the instantaneous NO reaction rate and an assumed beta PDF. The instantaneous NO reaction rate has contributions from thermal and prompt sources. The thermal pathway is modeled by the extended

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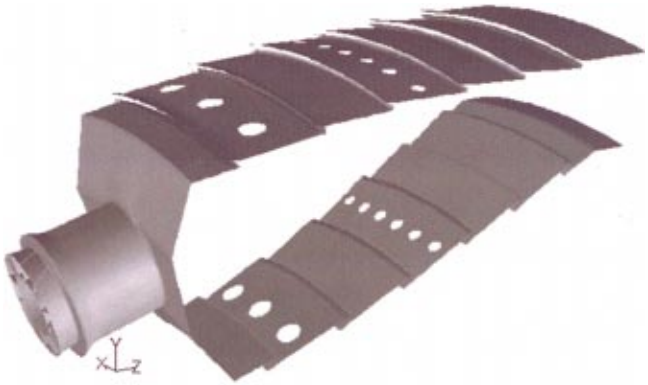

Fig. 1 Computational domain—20 deg annular secton of LM1600 combustion chamber

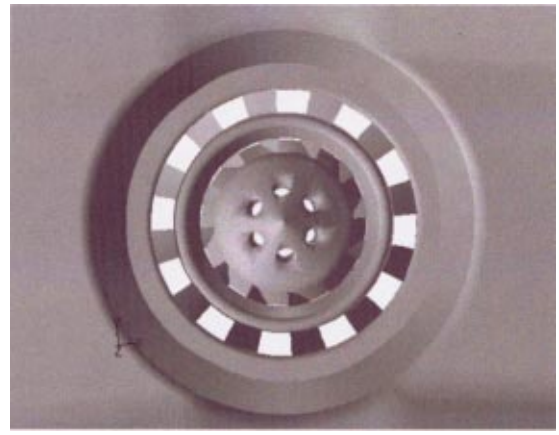

Fig. 2 Close up of the fuel injection port, primary air swirl vanes, and venturi nozzle details

Zeldovich mechanism with the steady-state assumption for the nitrogen radical. The prompt pathway is modeled via the empirical De Soete mechanism ([17]). Owing to the relatively high temperatures in the combustor, the thermal pathway is the predominant mechanism for NO formation.

Geometry, Grid Generation, and Boundary Conditions. The LM1600 gas turbine installed at the TransCanada compressor station is rated for 13.6 MW at standard site conditions. The annular combustion module consists of 18 fuel injection nozzles equally spaced about the periphery. The inherent geometric symmetry of the combustor was exploited by considering a 20-deg sector with a single fuel injection nozzle for the computational domain and employing periodic boundary conditions. The fuel port consists of a tube capped by a conical head with six small holes. Primary air enters the combustion chamber through a concentric annulus surrounding the fuel port. The primary air is swirled by twelve, 30 deg swirl vanes in the annulus before passing through a venturi nozzle and subsequently mixing with the natural gas jets. Dilution air is introduced further downstream through a series of holes and slots located along the top and bottom wall of the combustion liner. The combusted gases exit through an annular outlet, which leads to the multistaged high-pressure turbine. An overview of the 20-deg section of combustion chamber geometry is shown in Fig. 1. A close up front view of the fuel injection nozzle assembly, i.e., fuel injection port, primary air swirl vanes and venturi nozzle, is shown in Fig. 2.

A block-structured grid consisting of about 286,000 hexahedral control volumes with body-fitted coordinates was generated for the three-dimensional computational geometry. The boundary surfaces were uniquely identified during this process for later specification of boundary conditions. Second-order upwind differencing was employed for the convective terms in the discretized conservation equations in order to minimize numerical diffusion and achieve reasonably accurate solutions for this mesh size. Although a comprehensive grid refinement study was not performed, tests with coarser and slightly finer meshes showed only minor changes in predicted results. Unfortunately, complete grid independence is generally beyond available computing resources for most three-dimensional combustion problems.

The boundary conditions for the CFD simulations include periodic boundaries to connect the azimuthally rotated side planes, a constant pressure outlet, thin walls for the swirl vanes, and finally adiabatic, hydraulically smooth walls for the external surfaces such as the combustion liner and venturi nozzle. The mass flow rate of fuel, which varies with unit load, was specified from direct measurements with a conventional orifice meter.

Specification of the airflow boundary conditions requires some additional explanation, however, because the flow rate of air is not directly measured by the on-line data acquisition system that monitors the engine performance. Single point measurements of the exhaust gas velocity with a Pitot tube during the field tests did not provide reasonable estimates of the airflow. Instead, the air flow rate was indirectly determined by dividing the reported power output by the enthalpy change across the power turbine; the enthalpy change being derived from the measured pressure and temperature at the unit inlet and outlet. The actual breakdown of the total air flow rate into prescribed mass flow rates at the primary air inlet and the various secondary air slots and holes, shown in Fig. 1, was based on proprietary information supplied by General Electric.

Field Measurements

Field measurements were taken in order to determine the variation of emission levels from an LM1600 at various loads, i.e., engine power output. The tests were performed at the same TransCanada compressor station on two occasions, separated by almost one year. The ambient conditions at the site on these dates were quite similar: the air temperature for the first test ranged from 6–9°C compared to 11–13°C for the second test, while the barometric pressures remained within 1 percent.

The levels of NO_x, NO, CO, CO_2, O_2 and moisture in the exhaust stack were measured by a CEMS using instruments and on-site calibration techniques approved by U.S. EPA. For example, the NO_x measurements were made with an approved chemiluminescent continuous analyzer. Pertinent engine performance parameters such as the unit power output, fuel flow rate, turbine stage pressures and temperatures, and ambient conditions at the site were recorded by the data acquisition system at the station. Gas samples were also taken in order to determine the composition of the fuel gas. The results of the gas analysis are summarized below in Table I.

At each operating point, the unit speed was adjusted to reach the desired power output. Then the unit was allowed to stabilize before the CEMS measurements were taken over a 20-minute pe-

Table 1 Natural gas composition

Component	Field Test #1 Mole %	Field Test #2 Mole %
CH_4	96.92	96.75
C_2H_6	0.86	0.80
C_3H_8+	0.30	0.38
N_2	1.03	1.12
CO_2	0.89	0.95

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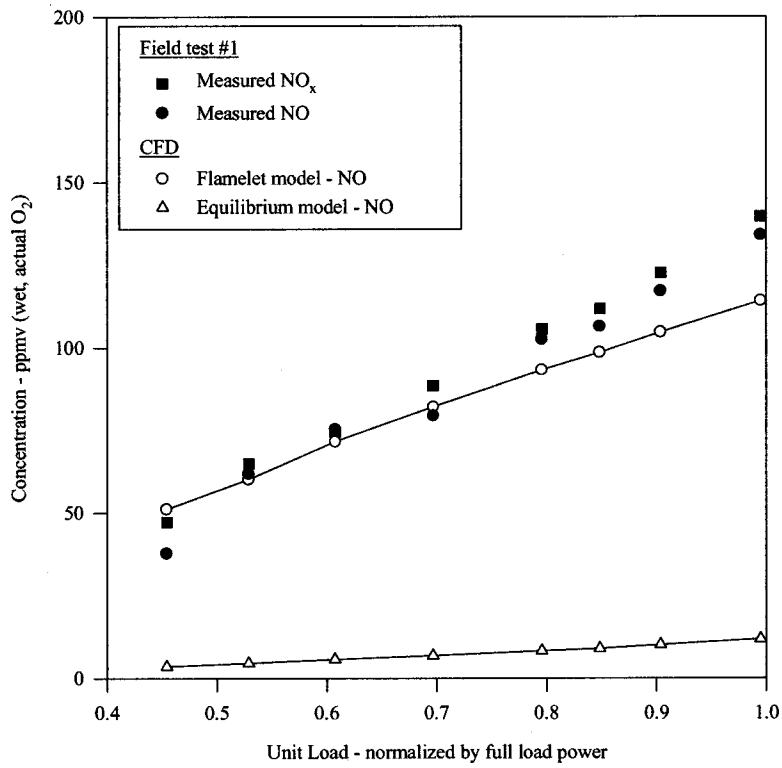

Fig. 3 Comparison of predicted (NO) with first set of CEMS based field measurements

riod. The unit load was varied from roughly 40 percent to maximum load to give eight operating points on each test date, for a total of 16 points.

Results and Discussion

CFD simulations were performed at each of the 16 operating points corresponding to the field measurements. At each operating point, multiple laminar flamelet files were generated to cover the range of strain rates in the combustor. Property look-up tables were also generated for both combustion models at each operating point. This was necessary because the absolute pressure and incoming compressed air temperature vary directly with the air compression ratio, which declines with decreasing unit load. Due to the extremely high methane content of the fuel gas, ~97 per-

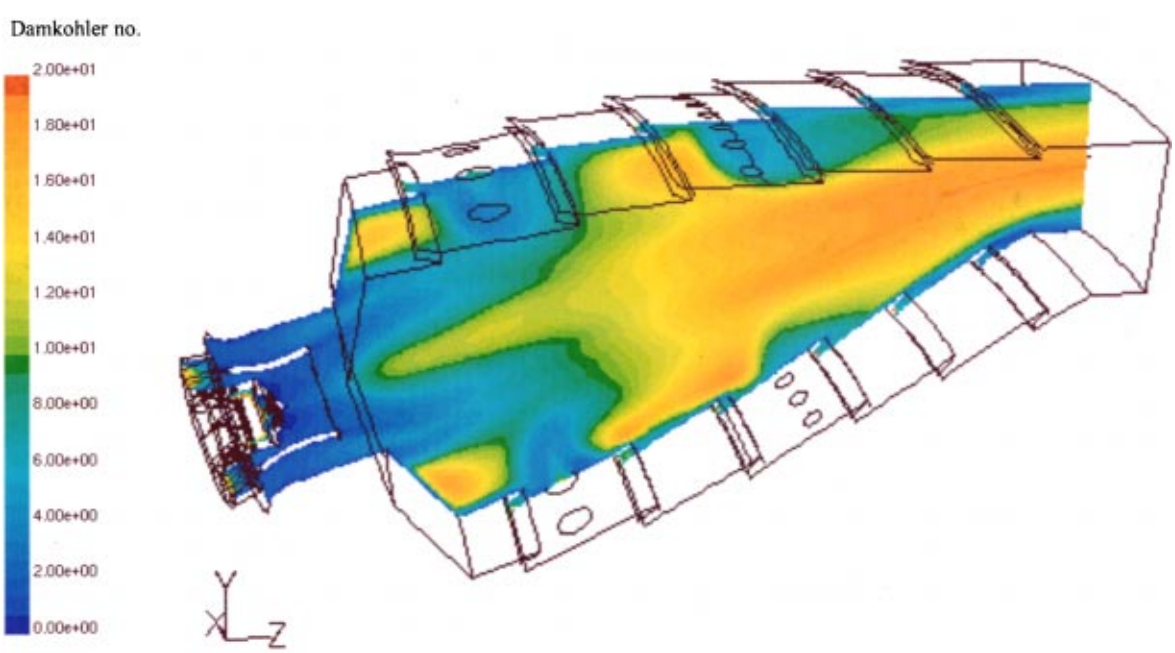

Fig. 4 Contours of the local Damkohler number on the central plane

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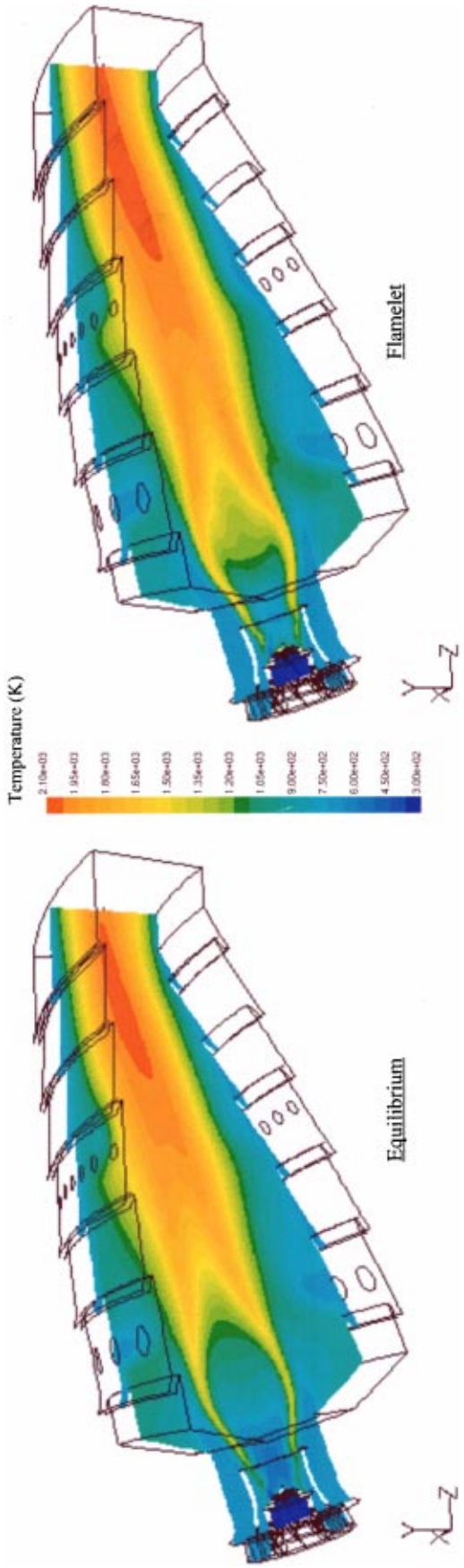

Fig. 5 Contours of static temperature on the central plane

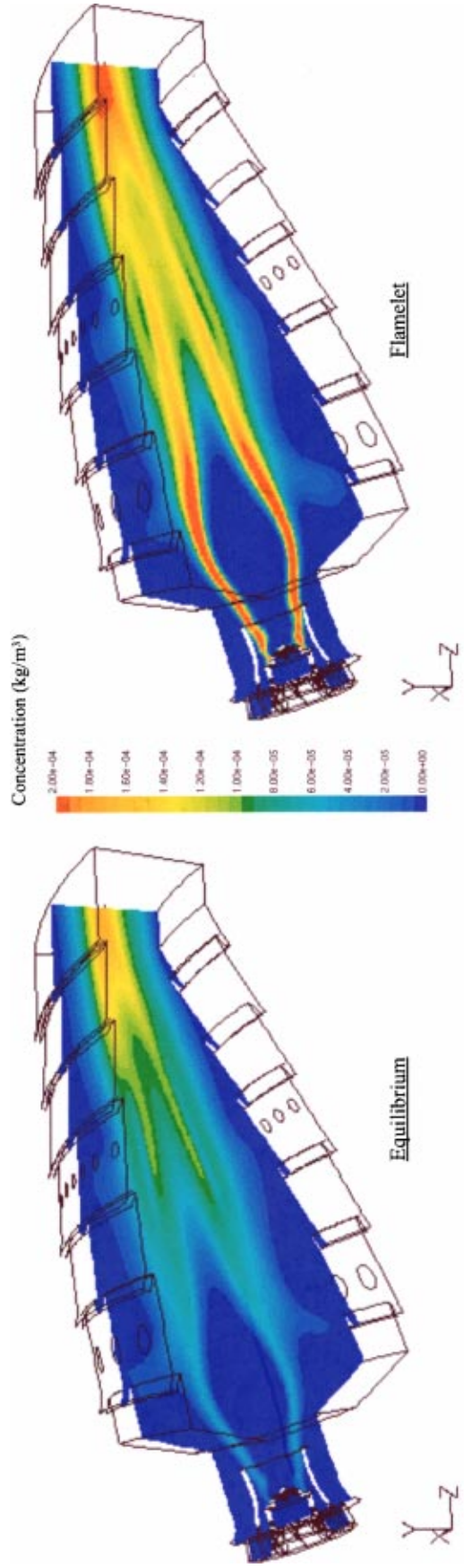

Fig. 6 Contours of oxygen radical concentration on the central plane

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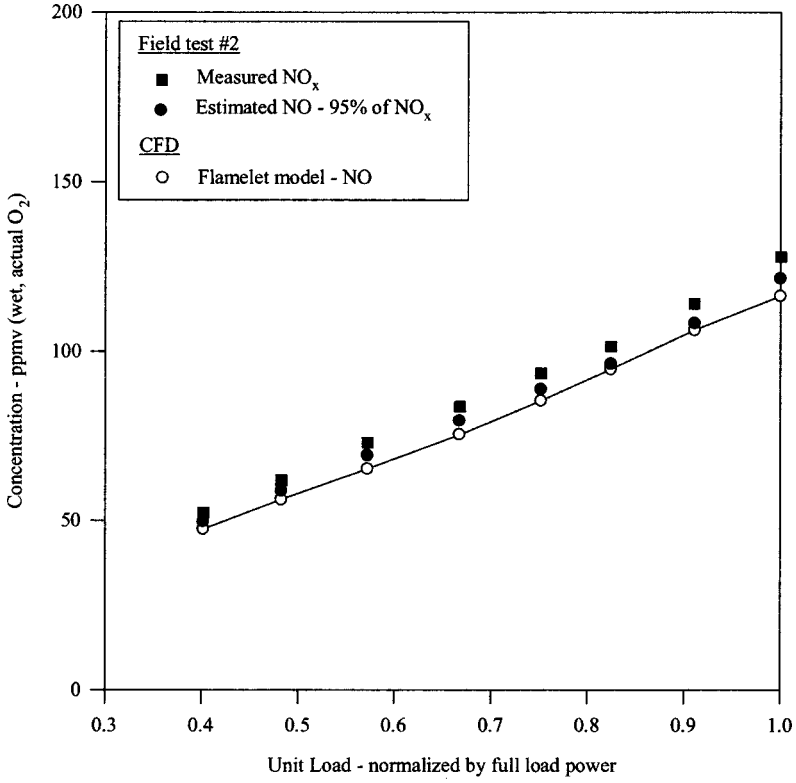

Fig. 7 Comparison of predicted (NO) with second set of CEMS based field measurements

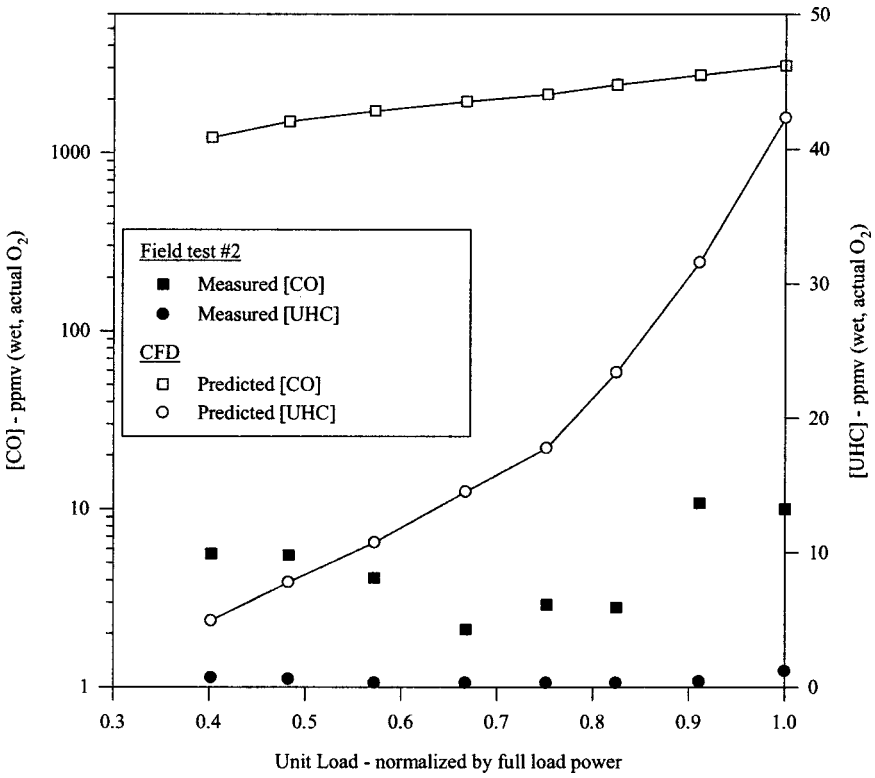

Fig. 8 Comparison of predicted (CO) and (UHC) with second set of CEMS based field measurements

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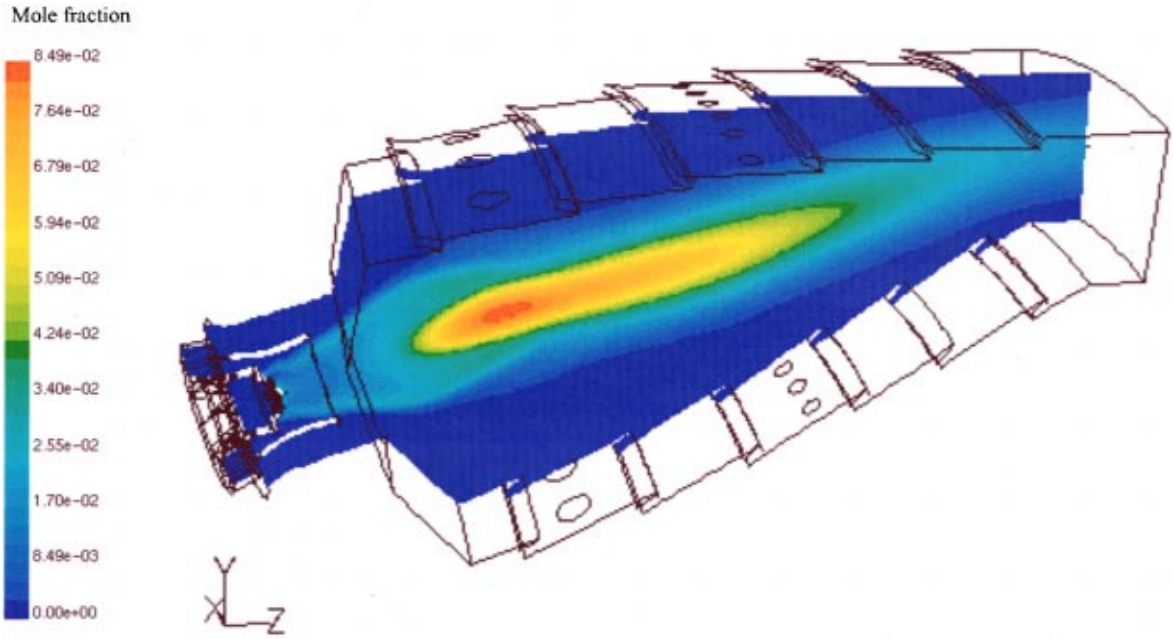

Fig. 9 Contours of CO mole fraction on the central plane

cent in Table 1, the reduced GRI-MECH 1.22 combustion mechanism ([20]) was implemented in the present analysis. This simplified methane mechanism considers 104 reactions and 22 species.

The predicted concentrations of NO from CFD were obtained by dividing the integrated volumetric flow rate of NO by the total integrated volumetric flow rate over the combustor outlet plane for the two different combustion models. The predicted concentrations were subsequently corrected to account for additional cooling air that bypasses the combustion chamber. These predictions are compared with the first set of CEMS measurements in Fig. 3. Note that NO was also measured during this first test along with the total NO_x to allow direct comparison with the CFD results. With the exception of two points in Fig. 3, NO accounts for about 95 percent of the total NO_x produced. The flamelet model results are clearly superior to the chemical equilibrium model, which under predicts the NO concentrations by an order of magnitude. The flamelet model only under predicts the NO concentrations by 15 percent at full load. Although this discrepancy diminishes at part loads, the rate of increase for the measured NO with increasing load is somewhat steeper than the predicted trend. The overall trend of increasing NO with increasing load is due to the higher temperatures in the combustor, which leads to additional NO via the thermal mechanism. Note these higher combustor temperatures result from higher compressed air temperatures and lower air-to-fuel ratios as the unit load is increased.

The inadequacy of the equilibrium based CFD model in this situation can be elucidated by considering the deviation from chemical equilibrium as measured by the Damkohler number, Da, which is defined as the ratio of characteristic turbulent time scale, k/ε, to chemical time scale, $1/b_s$. Here b_s is the laminar strain rate of a methane flame at extinction, $b_s = 11{,}700\ \mathrm{s}^{-1}$. Contours of the local Damkohler number are plotted on the central plane, i.e., vertical plane dissecting the computational domain, in Fig. 4. Ideally, this dimensionless parameter should be much greater than unity, or in other words the turbulent time scale should be much longer than the chemical time scale, for chemical equilibrium to be valid. This is clearly not the case in Fig. 4, where $5 < \mathrm{Da} < 20$ over a majority of the combustion chamber. Hence, there is insufficient time for chemical equilibrium to prevail.

Further insight into this problem is gained by comparing the temperature and oxygen radical concentrations for these two combustion models. In Fig. 5, contours of temperature on the central plane show only slight deviations between the two cases, although the high temperature region is larger for the flamelet model. The primary difference lies in the predicted concentration of oxygen radical [O] as shown in Fig. 6. Here, the flamelet model yields significantly higher [O] than predicted by equilibrium chemistry, particularly in the thin flame region near the injection nozzle. Higher [O] increase the thermal NO formation rate, which is directly proportional to the concentration of oxygen radial as shown by Mueller and Knill ([14]).

CFD simulations based on the second set of field measurements were performed only with the flamelet model. The integrated [NO] at the combustor outlet are compared with the average CEMS measurements in Fig. 7. Unfortunately, nitric oxide measurements were unavailable for this set of tests, so the reported [NO] in Fig. 7 was estimated at 95 percent of the measured NO_x based on the results of our previous set of field tests. Again, flamelet based CFD results slightly underpredict the measured trend but it is within 10 percent of the measured NO_x and within 5 percent of the estimated NO. This level of agreement is surprisingly good and the discrepancy can easily be attributed to additional NO formation via residence time during the time the gas leaves the combustion chamber and its subsequent expansion in the high-pressure turbine stage.

We also compared the CFD predictions of the unburned hydrocarbons (UHC) and carbon monoxide at the combustor outlet with the CEMS field measurements for the second set of tests in Fig. 8. Clearly the agreement between the predicted and measured values of UHC and CO is poor, particular for CO where the predicted levels are almost three orders of magnitude higher than the measurements. The inability of CFD to accurately predict these components was recognized by Nicol et al. ([21]).

Direct measurements of the [CO] and [UHC] at the outlet of a laboratory combustor chamber were reasonably well predicted by a flamelet-based model in Ref. [14]. Hence, we offer that these order of magnitude discrepancies can be explained by the discrepancy in the locations from which the CFD and measured values are derived. From Fig. 9, which shows the CO mole fraction, it appears that CO is produced in the fuel rich region of the flame. Further downstream in the fuel lean zone where the temperatures are higher (refer to Fig. 5) it is oxidized into CO_2. This oxidiza-

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tion process is relatively slow and will continue beyond the combustor, through the downstream turbine and even into the exhaust stack, where the pressure is atmospheric and the temperature ranges from 420–492°C depending on the unit load. Chemical equilibrium calculations show that negligible amounts of CO would be present under these conditions and the residence time to reach the measurement location on the exhaust stack is on the order of several seconds versus milliseconds in the combustor. Of course, better mixing could also reduce the UHC and CO emissions in the actual unit.

One last comment, the mass flow rate of air is an important parameter in the NO predictions but the indirect calculation method used to derive it is prone to inaccuracies due to uncertainties in the temperature measurements across the turbine. This is particularly true for an engine with a single turbine because the Turbine Inlet Temperature (TIT) is notoriously difficult to measure accurately due to hot/cold streaks in the flow from the combustion chamber. The LM1600 consists of two separate turbines: a high-pressure turbine to drive the air compression and a low-pressure turbine for power generation. Fortunately, the temperature measurements across the low-pressure turbine are significantly more accurate than the TIT and these are used in the air flow calculation. An additional CFD simulation at full load conditions revealed that a 10 percent increase in the total air flow rate reduced the predicted [NO] by 13 percent. This decrease is largely due to effect of increased air dilution along with slightly enhanced mixing in the combustion chamber. The sensitivity analysis also provides a gross estimate of the uncertainty in [NO] associated with the inaccuracy in the calculated flow rate of air through the combustion chamber.

Concluding Remarks

The following conclusions can be drawn from the present study:

1 The predicted [NO] with the flamelet model was in much closer agreement with CEMS field measurements than the chemical equilibrium model. This is likely because it more accurately predicts the [O] within the combustor, which is an important factor in thermal NO mechanism.

2 Even with all the extra details and efforts incorporated in the flamelet model, the best CFD predictions were shown to be within 15 percent at maximum load.

3 The field measurements showed that NO constitutes about 95 percent of the total NO_x measured in this type of engine.

4 UHC and CO emissions were greatly overpredicted, arguably as a result of neglecting their post-combustor oxidation in the high-temperature fuel-lean environment of the turbine and exhaust stack.

Acknowledgments

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Nomenclature

a = absorption coefficient
b_s = laminar flamelet strain rate at extinction
c_p = specific heat
D = diffusion coefficient
Da = Damkohler number, $\mathrm{Da}=b_s k/\varepsilon$
f = mixture fraction
k = turbulent kinetic energy
Le = Lewis number, $\mathrm{Le}=\rho c_p D/\lambda$
p = static pressure
P = probability density function
T = static temperature
u_i = velocity components, $i=1,2,3$
$\dot{w}$ = reaction rate
X = mole fraction
Y = mass fraction

Greek Symbols

χ = scalar dissipation rate
ε = turbulence dissipation rate
λ = thermal conductivity
ρ = fluid density
σ = Stefan-Boltzmann constant

Subscripts

b = background conditions
j = current species

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